Seamoon Pandey

B.E. Computer Engineering (2026) | AI & Backend Engineer

pandeyseamoon05@gmail.com | +977 9866142928 | github.com/seamoonpandey | linkedin.com/in/seamoonpandey

Tansen-14, Palpa, Nepal | Date of birth: 27 January 2003 | Nationality: Nepali

PROFILE

2026 Computer Engineering graduate of the Institute of Engineering, Tribhuvan University, applying for an M.Sc. in Computer Science. My final-year thesis built a cross-site scripting scanner on a fine-tuned DistilBERT classifier and a microservice backend. I now lead the Bitfolio team at Lamida Inc., building data systems for the Bittensor network.

EDUCATION

Pashchimanchal (Western Region) Campus, Institute of Engineering, Tribhuvan University — Pokhara, Nepal

Bachelor of Engineering in Computer Engineering — 2022 – 2026 (graduated)

Aggregate: **67.98%**, **First Division** | 1096.1 / 1612.5 marks | T.U. Regd. 3-2-45-173-2021

Tribhuvan University scale: Distinction $\geq$ 80%, First Division $\geq$ 65%, Second Division $\geq$ 50%. Examinations are set and moderated centrally by the university and marks are not scaled.

Thesis: RedSentinel, an AI-assisted cross-site scripting (XSS) vulnerability scanner. Lead architect, team of 3.

LANGUAGE PROFICIENCY

- **English:** IELTS Academic **7.5** overall, **CEFR C1** (15 August 2026). Listening 8.5 | Reading 7.5 | Writing 7.0 | Speaking 7.0. No band below 7.0.
- **Nepali:** native speaker. **Medium of instruction:** English, throughout the Bachelor's degree.

THESIS PROJECT

RedSentinel: AI-Assisted XSS Vulnerability Scanner — Final-year Thesis, Team of 3 | Code | Report | Demo

Lead architect; owned the NestJS orchestration core and the end-to-end ML pipeline. Teammates built the fuzzer and obfuscation engines.

- Fine-tuned a **DistilBERT** multi-task classifier to label an injection point's context and severity, replacing brittle signature matching, and reached near-perfect accuracy on an **8,869-sample** held-out split of a deterministic, well-structured label space.
- Built the full ML pipeline on a **59,122-entry** curated XSS dataset: data curation, custom tokenizer training, multi-task training with checkpointing, evaluation, and an inference API with ONNX/TorchScript export and confidence calibration.
- Architected a polyglot microservice system: a **TypeScript**/**NestJS** core orchestrating three **Python**/**FastAPI** AI services over a **Redis**/**BullMQ** job queue, with **PostgreSQL** persistence and a real-time **Next.js** dashboard (Socket.IO).
- Co-developed an **XGBoost** payload ranker (35 engineered features) to prioritise payloads by predicted effectiveness, with a heuristic fallback when the trained artifact is unavailable.

EXPERIENCE

Team Lead and Full-Stack Engineer, Lamida Inc. — Full-time | 23 Feb 2026 – Present

- **Team Lead, Bitfolio** from September 2026: market intelligence and paper trading for subnets of **Bittensor**, a decentralised machine-learning network. Its **Python** collector ingests X, Discord, taostats and on-chain data across 129 subnets.
- Found a silent production data fault by auditing stored rows, not row counts: an upstream API ignored its subnet filter and a duplicate guard kept the first bad response, so whale signals used another subnet's holders for weeks. Moved holder data to **on-chain metagraph reads**, added rename and coldkey-swap history, and purged **73,250** corrupt rows.
- Build full-stack **Node.js** applications for Web3 products, including a DAO for the tUSDT token.

Freelance Full-Stack Developer — Contract, Remote

- Delivered production platforms on **Express.js + React** (auth, REST APIs, third-party integrations) while carrying a full course load: **Jyotisam**, an astrology booking platform (jyotisam.com), and **Mahadaan**, a donation platform (mahadaan.com).

Mentor, Call for Enthusiasts Programme — Innovative Computer Engineering Students' Society (i-CES)

- Mentored junior students on scalable application design, clean backend architecture, and development practice.

TECHNICAL SKILLS

- **Languages:** Python, TypeScript, JavaScript, Ruby, C/C++, Dart
- **AI / ML:** PyTorch, Transformers (DistilBERT), XGBoost, YOLOv8, ONNX/TorchScript, feature engineering, model evaluation
- **Backend:** Node.js, NestJS, FastAPI, Express.js, Flask, Ruby on Rails, REST, WebSocket/Socket.IO
- **Data & Infra:** PostgreSQL, SQLite, Redis, BullMQ, Docker Compose, GitHub Actions CI/CD, systemd, job queues
- **Frontend & Security:** React, Next.js, Flutter; XSS detection, fuzzing, WAF analysis, headless-browser verification (Playwright)
- **Practices:** system architecture, MVC, auth (JWT/API keys), e2e/unit testing, Git, Linux

SELECTED PROJECTS

- **AI Artifact Recognition Platform** (PEC Hackathon: **YOLOv8**, Flask, React). Real-time object recognition with async inference.
- **Community Connect** (OSM Hackathon: Express.js, Flutter). Event-coordination platform with GPS-based attendance validation.
- **Garva Diary** (BMC InnovateX: Ruby on Rails). Rails MVC backend APIs with validation workflows and mobile integration.